

MODEL THEORY

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1. INTRODUCTION

Model theory is a part of logic where one studies classes of models of various sets of axioms. The fundamental questions of the area are: how to classify/construct models, what configurations can appear in a model, and how are different theories (sets of axioms) connected to each other. The standard methods of model theory are set up in the framework of first-order logic, i.e., one can only take quantifiers over elements, not over subsets. In applications it is often a non-trivial task to translate a problem into a first-order equivalent, but when possible it can lead to strong results.

Every theory is expressed in a language, e.g., groups have group multiplication, rings have $+$, $\cdot$, and vector spaces over a field have $+$ and multiplications by scalars. Equality is assumed to be in the language by default.

Example 1.1. Consider the theory

$$T = \left\{ (\exists x_1, \dots, x_n) \left(\bigwedge_{i \neq j} x_i \neq x_j \right) : n \in \mathbb{N} \right\}.$$

Models of T are just infinite sets $M \models T$. Up to isomorphism, these are classified by their cardinality. For a formula

$$\varphi(\bar{x}) = \exists y_1 \forall y_2 \dots \exists y_{2n-1} \forall y_{2n} \bigwedge_i \bigvee_j (\neg) \alpha_{ij} = \beta_{ij},$$

where $\alpha_{ij}, \beta_{ij} \in \{x_a, y_b : a, b\}$, one can show ⑥ that $\varphi(M) \subseteq M^{|\bar{x}|}$ is a Boolean combination of weak diagonals.

Of course, some properties of infinite sets depend on the model of ZFC we are working with, but we will treat the cardinal hierarchy as absolute, and rather reduce questions about models to, for example, cardinality questions. Surprisingly, many properties of the class of models of a theory are independent of the model of ZFC we choose. For a discussion on why we are working with a metatheory, see [3, 4].

For example, regardless of the universe of set theory, for the above theory T its spectrum function

$$I(T, \kappa) := |\{M \models T : |M| = \kappa\}| / \text{iso.}|$$

is constantly equal to 1. A theory T such that $I(T, \kappa) = 1$ is called κ -categorical, and characterisation of such theories (for $\kappa = \aleph_0$ or $\kappa > \aleph_0$) is one of the most fundamental results in model theory (Ryll-Nardzewski's and Morley's theorem, respectively).

The scope of model theory has increased substantially since its origin. The bounds of the area are ultimately defined by what its users can prove. Nonetheless, there are several slogans describing what model theory is, that perhaps reflect the history of the development of the subject:

[Chang and Keisler, 1973] Model theory = Logic + Universal algebra,

[Hodges, 1997] Model theory = Algebraic geometry - Fields,

[Hrushovski] Model theory is the geography of tame mathematics.

These will hopefully become more meaningful during the course-time. The term “geography” in Hrushovski's description may be loosely treated as a reference to Shelah's idea of drawing division lines on the map of all theories: defining properties of theories that reflect the structure of their models. This point of view is exemplified quite literally in the model theoretic “map of the universe” [2].

Each ‘geographical’ region has a different culture and can lead to different interesting applications. During the course we can venture into various model-theoretic continents including frameworks for:

- geometry of differential equations and differential Galois theory (stability);
- difference equations, pseudo-finite fields (simplicity);
- tame topology and real-algebraic geometry (o-minimality);
- valued fields, tame non-Archimedean geometry (NIP, generic stability).

Depending on the interests of the audience we will choose one or more directions and from time to time we will supplement examples from the corresponding regions.

2. COMPLETENESS THEOREM

The fundamental tool of model theory is Gödel's completeness theorem. It was proven by Kurt Gödel in his PhD thesis in 1929 and reformulated in Leon Henkin's PhD thesis in 1947. The theorem says that every first-order statement that is true in every model of some theory T admits a proof from the axioms of T . To state the completeness theorem formally we need to set the stage.

Definition 2.1. A language L consists of collections of *function symbols* $\{f_i\}_{i \in I}$, *relation symbols* $\{P_j\}_{j \in J}$, and *constant symbols* $\{c_k\}_{k \in K}$. Furthermore, each function and relation symbol is equipped with its *arity*, which is a natural number n_i for $i \in I$, m_j for $j \in J$.

Fix a language L for the rest of this section.

Definition 2.2. An L -structure is a non-empty set M together with *interpretations* of:

- functions $f_i^M : M^{n_i} \rightarrow M$ for $i \in I$,
- relations $P_j^M \subseteq M^{m_j}$ for $j \in J$,
- constants $c_k^M \in M$ for $k \in K$.

Definition 2.3. Fix an infinite set of variables $\{v\}_{v \in V}$. The L -terms (in variables $\{v\}_{v \in V}$) are elements of the smallest set that contains the variables, constant symbols, and whenever $i \in I$, $t_1, \dots, t_{n_i}$ are terms, then so is $f_i(t_1, \dots, t_{n_i})$.

We define the set of L -formulas (in variables $\{v\}_{v \in V}$) inductively as follows:

- An *atomic formula* is either of the form $t_1 = t_2$, where t_1, t_2 are terms in variables $\{v\}_{v \in V}$, or of the form $P_j(t_1, \dots, t_{m_j})$ for some $j \in J$ and terms $t_1, \dots, t_{m_j}$.
- If φ is a formula, then so is $\neg\varphi$.
- If φ, ψ are formulas, then so are $\varphi \wedge \psi, \varphi \vee \psi$.
- If φ is a formula and $v \in V$, then so are $(\exists v)(\varphi)$ and $(\forall v)(\varphi)$.

It would be enough to use e.g. $\wedge, \neg, \exists$ as logical symbols, and define other ones as combinations of those, for example $(\forall x)(\varphi) := \neg(\exists x)(\neg\varphi)$. Similarly, we use the conventions $(\varphi \vee \psi) := \neg(\neg\varphi \wedge \neg\psi)$, $(\varphi \rightarrow \psi) := (\neg\varphi \vee \psi)$ and $(\varphi \leftrightarrow \psi) := (\varphi \rightarrow \psi) \wedge (\psi \rightarrow \varphi)$. Also, we write $\bigwedge_{i=1}^l \varphi_i, \bigvee_{i=1}^l \varphi_i$ for the inductively defined iterations of $\wedge, \vee$ respectively. We often skip parentheses, L and variables in the names, if those are clear from the context. We fix an infinite set of variables $\{v\}_{v \in V}$ for the rest of this section.

For every term t and formula φ we inductively define the set of variables $FV(t)$, $FV(\varphi)$ occurring freely in t , φ respectively, so that:

- (i) $FV(v) = \{v\}, FV(c_k) = \emptyset, FV(f_i(t_1, \dots, t_{n_i})) = \bigcup_{a=1}^{n_i} FV(t_a)$.
- (ii) $FV(\neg\varphi) = FV(\varphi), FV(\varphi \wedge \psi) = FV(\varphi) \cup FV(\psi)$.
- (iii) $FV(\exists v \varphi) = FV(\varphi) \setminus \{v\}$.

We often denote variables by $x_1, \dots, x_n$. We write $t(x_1, \dots, x_n)$ or $\varphi(x_1, \dots, x_n)$ to mean that the free variables of t or φ respectively, are among $x_1, \dots, x_n$. We use overlines as tuples of variables $\bar{x} = (x_1, \dots, x_n)$.

Definition 2.4. A formula without free variables is called a *sentence*. A set of sentences is called an L -theory.

If M is an L -structure and $t(x_1, \dots, x_n)$ is a term, we inductively define the interpretation of t as a function $t^M : M^n \rightarrow M$, or as an element of M if $FV(t) = \emptyset$. If t is a single variable or a constant c_k , t^M is the appropriate projection function $M^n \rightarrow M$ or $c_k^M \in M$ respectively. If $t = f_i(t_1, \dots, t_{n_i})$, then t^M is defined by applying f_i^M to the values of $t_1^M, \dots, t_{n_i}^M$ on the appropriate tuple of elements of M . The precise definition of this step is left as an exercise ⑥.

Definition 2.5 (Tarski's definition of truth). Let M be an L -structure. We define a relation $M \models \varphi(a_1, \dots, a_n)$ for formulas $\varphi(x_1, \dots, x_n)$ and (possibly empty) tuples $\bar{a} = (a_1, \dots, a_n) \in M^n$ inductively as follows:

- (1) If $\varphi(\bar{x})$ is $t_1(\bar{x}) = t_2(\bar{x})$, then $M \models \varphi(\bar{a})$ if $t_1^M(\bar{a}) = t_2^M(\bar{a})$.
- (2) If $\varphi(\bar{x})$ is $P_j(t_1(\bar{x}), \dots, t_{m_j}(\bar{x}))$ then $M \models \varphi(\bar{a})$ if $(t_1^M(\bar{a}), \dots, t_{m_j}^M(\bar{a})) \in P_j^M$.
- (3) $M \models \neg\varphi(\bar{a})$ if it is not the case that $M \models \varphi(\bar{a})$ (written $M \not\models \varphi(\bar{a})$).
- (4) $M \models (\varphi \wedge \psi)(\bar{a})$ if $M \models \varphi(\bar{a})$ and $M \models \psi(\bar{a})$.

- (5) If $\varphi(\bar{x}) = (\exists y)(\psi(\bar{x}, y))$, then $M \models \varphi(\bar{a})$ if there is $b \in M$ such that $M \models \psi(\bar{a}, b)$.

If $M \models \varphi(\bar{a})$, we say that M *models* $\varphi(\bar{a})$. If ψ has free variables $x_1, \dots, x_n$, by convention we define $M \models \psi$ to mean $M \models \bar{\psi}$, where $\bar{\psi} = \forall x_1 \dots \forall x_n \psi$ is the *universal closure* of ψ . For an L -theory T we say that an L -structure M is a *model* of T , written $M \models T$, if for every $\varphi \in T$, $M \models \varphi$.

In order to state Gödel's completeness theorem it remains to define the notion of a proof. First, recall that a *propositional formula* is an expression built using propositional symbols $P_1, P_2, \dots$ and Boolean connectives $\neg, \wedge$ (we can also add $\vee, \rightarrow, \leftrightarrow$, or define them similarly as before). We say that a propositional formula is a *tautology*, if under any substitution of *Truth* and *False* for P_i 's, the resulting term in the Boolean algebra $\{\text{Truth}, \text{False}\}$ is evaluated as *Truth*. If $\tau(P_1, \dots, P_n)$ is a propositional formula, and $\varphi_1, \dots, \varphi_n$ are formulas, we call $\tau(\varphi_1, \dots, \varphi_n)$ an *instantiation* of τ .

Lemma 2.6. If $\tau(P_1, \dots, P_n)$ is a tautology and $\varphi_1, \dots, \varphi_n$ are formulas, then for all L -structures M we have

$$M \models \tau(\varphi_1, \dots, \varphi_n).$$

Proof. Exercise ©. □

Definition 2.7. The *axioms of the classical first-order logic* (in the language L and variables $\{v\}_{v \in V}$) consist of:

- A0. Instantiations $\tau(\varphi_1, \dots, \varphi_n)$, for tautologies τ and formulas $\varphi_1, \dots, \varphi_n$.
- A1. Formulas $\forall x(\varphi \rightarrow \psi) \rightarrow (\varphi \rightarrow \forall x \psi)$, for all pairs of formulas φ, ψ , such that x is not free in φ .
- A2. Formulas $\forall x \varphi \rightarrow \varphi(x/t)$, for all formulas φ and terms t such that none of the free occurrences of x in φ is in the range of a quantifier bounding a free variable from t . Here $\varphi(x/t)$ is the result of substitution of every free occurrence of x in φ by t .
- A3. Equality axioms:
 - $x = x$ for every variable x ;
 - $x = y \rightarrow t(\dots, x, \dots) = t(\dots, y, \dots)$, for every term t and every pair of variables x, y , where y appears in the same place in t as x was;
 - $x = y \rightarrow (\varphi(\dots, x, \dots) \leftrightarrow \varphi(\dots, y, \dots))$, for every formula φ and every pair of variables x, y , where y appears in the same place in φ as x was, and x occurs freely in φ , not in the range of a quantifier bounding y .

To be extremely formal in the above definition we would have to define $\varphi(x/t)$ and conditions like “none of the occurrences of x in φ is in the range of a quantifier bounding a free variable from t ” inductively. We leave it as an exercise © to the reader. If $\varphi(\bar{x})$ is a formula and $\bar{c}$ is tuple of constants of the same length, then we write $\varphi(\bar{c})$ for the substitution of respective x_i 's by c_i 's. Note that the restriction in A2 makes sense, as if $\varphi = (\exists y)(x \neq y)$ and $t = y$, then it should not be an axiom that

$$\forall x \exists y x \neq y \rightarrow (\exists y)(y \neq y).$$

Definition 2.8. The *deduction rules of the classical first-order logic* consist of the modus ponens (MP) rule

$$\frac{\varphi, \varphi \rightarrow \psi}{\psi}$$

and the $\forall$ -rule

$$\frac{\varphi}{\forall x \varphi},$$

for every pair of formulas φ, ψ .

Definition 2.9. We say that a set of formulas X proves a formula ψ , written $X \vdash \psi$, if there is a finite sequence of formulas $\alpha_1, \dots, \alpha_n = \psi$, such that for every $i = 1, \dots, n$ one of the following two conditions holds:

- (i) α_i is an axiom of the classical first-order logic or an element of X .
- (ii) α_i can be deduced from $\alpha_1, \dots, \alpha_{i-1}$ using one of the deduction rules. More precisely, there are $j, k < i$ such that $\alpha_k = (\alpha_j \rightarrow \alpha_i)$, or $\alpha_i = \forall x \alpha_k$.

If $X = \emptyset$ in the above definition, we just write $\vdash \psi$ instead of $\emptyset \vdash \psi$.

Example 2.10. Let us show that $\vdash \forall x \psi \rightarrow \exists x \psi$. Let β be the formula we want to prove and let y be a variable not appearing in β . The proof can be constructed as follows:

- α_1 : $\forall x \psi \rightarrow \psi(x/y)$, by A2.
- α_2 : $\forall x \neg \psi \rightarrow \neg \psi(x/y)$, by A2.
- α_3 : $\alpha_2 \rightarrow (\psi(x/y) \rightarrow \neg \forall x \neg \psi)$, by A0.
- α_4 : $\psi(x/y) \rightarrow \neg \forall x \neg \psi$, by MP.
- α_5 : $\alpha_1 \rightarrow (\alpha_4 \rightarrow \beta)$, by A0.
- α_6 : $\alpha_4 \rightarrow \beta$, by MP.
- α_7 : β by MP.

Exercise 2.11 (Deduction theorem). (e) Let φ be a sentence, ψ be a formula, and X be a set of formulas. Then:

$$X \vdash \varphi \rightarrow \psi \iff X \cup \{\varphi\} \vdash \psi.$$

Exercise 2.12. (e) For any formula φ we have $\varphi \vdash \bar{\varphi}$ and $\bar{\varphi} \vdash \varphi$.

For a formula φ and a theory T , we write $T \models \varphi$, if for every L -structure $M \models T$ we have $M \models \varphi$. If $T = \emptyset$ we abbreviate it as $\models \varphi$.

Theorem 2.13 (Soundness). Let φ be a formula. If $\vdash \varphi$, then $\models \varphi$.

Proof. By induction, left as an exercise. (e). □

Note that by the finitary nature of proofs and the deduction lemma, it follows that for a theory T , $T \vdash \psi$ implies that $T \models \psi$.

Theorem 2.14 (Gödel's completeness). Let φ be a formula. If $\models \varphi$, then $\vdash \varphi$.

A set of formulas S is *consistent*, if $S \not\vdash (\exists x)(x \neq x)$.

Theorem 2.15 (Completeness after Henkin). Let S be a consistent set of sentences. Then S has a model.

Proof. Let κ be the maximum of cardinalities of the number of variables and size of the language. Consider a new language L' which has the same symbols as L together with κ -many new constant symbols $(d_\lambda)_{\lambda < \kappa}$. Let $(\varphi_i(x))_{i < \kappa}$ be a sequence of L' -formulas with one free variable x . Let $g : \kappa \rightarrow \kappa$ be an increasing function such that for every $\lambda < \kappa$, $d_{g(\lambda)}$ does not appear in $\varphi_i(x)$ for $i \leq \lambda$. Now consider the set of L' -sentences

$$S' = S \cup \{\exists x \varphi_i(x) \rightarrow \varphi_i(d_{g(i)}) : i < \kappa\}.$$

We claim that S' is consistent. Indeed, if it were not, there would be $i_1 < \dots < i_n$ such that

$$S'' = S \cup \{\exists x \varphi_{i_j}(x) \rightarrow \varphi_{i_j}(d_{g(i_j)}) : j = 1, \dots, n-1\}$$

is consistent, and it becomes inconsistent if $\exists x \varphi_{i_n}(x) \rightarrow \varphi_{i_n}(d_{g(i_n)})$ is added to it. Thus, $S'' \vdash \exists x \varphi_{i_n}(x) \wedge \neg \varphi_{i_n}(d_{g(i_n)})$. In particular $S'' \vdash \neg \varphi(d_{g(i_n)})$. By the definition of $g(i_n)$, $d_{g(i_n)}$ does not appear in formulas from S'' . Hence, we can rewrite the proof witnessing that $S'' \vdash \neg \varphi_{i_n}(d_{g(i_n)})$, so that we get $S'' \vdash \neg \varphi_{i_n}(x)$. This contradicts the fact that $S'' \vdash \exists x \varphi_{i_n}(x)$.

Let T be a maximal consistent L' -theory containing S' . Such an extension exists by the Kuratowski-Zorn lemma. It is easy to see that for every L' -sentence φ , we have $\varphi \in T$ or $\neg \varphi \in T$ (theories with this property are called *complete*). We consider a relation on the set of constants $\{d_\lambda\}_{\lambda < \kappa}$ defined by

$$d_\lambda \sim d_\mu \iff T \vdash d_\lambda = d_\mu.$$

It is an equivalence relation and we define an L -structure M by considering the set of equivalence classes of $\sim$, and by requiring that:

- (1) $f^M([d_{i_1}]_\sim, \dots, [d_{i_n}]_\sim) = [d_j]_\sim \iff T \vdash f(d_{i_1}, \dots, d_{i_n}) = d_j$ for every function symbol f of arity n in L ;
- (2) $([d_{i_1}]_\sim, \dots, [d_{i_m}]_\sim) \in R^M \iff T \vdash R(d_{i_1}, \dots, d_{i_m})$ for every relation symbol R of arity m in L ;
- (3) $c^M = [d_j]_\sim \iff T \vdash c = d_j$, for every constant symbol c in L .

We leave the proof of the fact that these are well defined and do not depend on choices for the reader ©. The existence of the d_j in the cases (1), (3) follows from the fact that for every term t without free variables, in any language, $\vdash \exists x t = x$, which we leave as an exercise ©. It remains to show that $M \models \varphi$, for $\varphi \in S$. More generally, we will show that

$$M \models \alpha \iff T \vdash \alpha,$$

for all L' -sentences (not formulas), where we interpret d_i^M as $[d_i]_\sim$. We do it by induction on formula complexity of α :

- If α is an atomic L' -formula, then it must be of the form $R(t_1, \dots, t_m)$ or $t_1 = t_2$, for some terms $t_1, t_2, \dots, t_m$ and a relation symbol R of arity m . If $t_i^M = [d_{\lambda_i}]_\sim$ for all i , then $T \vdash t_i = d_{\lambda_i}$ for all i , so e.g.

$$\begin{aligned} M \models \alpha &\iff M \models R([d_{\lambda_1}]_\sim, \dots, [d_{\lambda_m}]_\sim) \\ &\iff T \vdash R(d_{\lambda_1}, \dots, d_{\lambda_m}) \iff T \vdash R(t_1, \dots, t_m). \end{aligned}$$

The case of equality of two terms is the same.

- If α is a conjunction then the equivalence is obvious. If $\alpha = \neg \beta$, then

$$M \models \alpha \iff M \not\models \beta \iff T \not\vdash \beta \iff T \vdash \neg \beta,$$

where the last equivalence uses the fact that T is maximal consistent.

- If $\alpha = \exists x \varphi_i(x)$, then we choose a witness $[d_\lambda]_\sim$ and we get

$$M \models \varphi_i([d_\lambda]_\sim) \iff M \models \varphi_i(d_\lambda) \iff T \vdash \varphi_i(d_\lambda) \implies T \vdash \exists x \varphi_i(x).$$

The middle equivalence follows from the inductive hypothesis. On the other hand

$$\begin{aligned} T \vdash \exists x \varphi_i(x) &\implies T \vdash \varphi_i(d_{g(i)}) \iff M \models \varphi_i(d_{g(i)}) \\ &\iff M \models \varphi_i([d_{g(i)}]_\sim) \implies M \models \exists x \varphi_i(x). \end{aligned}$$

Here the first equivalence follows from the inductive hypothesis.

This completes the proof, as $S \subseteq T$. $\square$

Henkin's proof of completeness is important as it provides a step-by-step 'construction'. It can be generalised to construct models with some specific non-first-order properties.

Corollary 2.16. $\textcircled{e}$ Let T be a theory and ψ be a sentence. Then

$$T \models \psi \iff T \vdash \psi.$$

Corollary 2.17 (Compactness theorem). $\textcircled{e}$ Let T be a theory. Then T has a model if and only if for every finite $T_0 \subseteq T$, T_0 has a model.

From now on, we identify theories that have the same logical consequences, i.e., when we write $T = T'$, we mean $T \vdash T'$ and $T' \vdash T$, or equivalently that for any L -structure M , $M \models T \iff M \models T'$.

Exercises, sheet 1.

1. Fill the gaps left in the proofs above. Exercises throughout the text are marked with the symbol $\textcircled{e}$. Read the proof of completeness theorem carefully.
2. Prove that if X is a set of formulas and $X \vdash \varphi(c)$, where c is a constant not appearing in any formula from X and $\varphi(x)$ is a formula with a free variable x , then $X \vdash \forall x \varphi(x)$.
3. Show that in any language φ there is a prenex normal form formula ξ such that $\vdash \varphi \leftrightarrow \xi$.
4. Let $L = (+, \cdot, <, (c_r)_{r \in \mathbb{R}}, \dots)$, where in (...) we include a unary (i.e., of arity one) function symbol for every function $f : \mathbb{R} \rightarrow \mathbb{R}$. Consider $\mathbb{R}$ as an L -structure. Show that there is a model M of $T = Th(\mathbb{R}) = \{\sigma : \mathbb{R} \models \sigma\}$ such that in M there exists $\varepsilon > 0$ with $\varepsilon < c_r^M$ for all $r \in \mathbb{R}_{>0}$. Hint: use compactness theorem for some $L' = L \cup \{\epsilon\}$ -theory and put $\varepsilon = \epsilon^M$.
5. Note that the model M from the previous exercise contains an isomorphic copy of $\mathbb{R}$ (even as an L -structure). We will identify this copy with $\mathbb{R}$ itself.
6. Let $\langle \mathbb{R} \rangle \subseteq M$ be the convex hull of $\mathbb{R} \subseteq M$. Note that $x \in \langle \mathbb{R} \rangle$ if and only if $-n < x < n$ for some $n \in \mathbb{N}$. Show that for any such $x \in \langle \mathbb{R} \rangle \subseteq M$ there is a unique $r \in \mathbb{R}$ such that for all $m \in \mathbb{N}$ we have $-1/m < x - r < 1/m$. We denote it by $r = \text{st}(x)$.
7. Prove that $\text{st} : \langle \mathbb{R} \rangle \rightarrow \mathbb{R}$ has the following properties: $\text{st}(x + y) = \text{st}(x) + \text{st}(y)$, $\text{st}(xy) = \text{st}(x)\text{st}(y)$. Let $g : \mathbb{R} \rightarrow \mathbb{R}$ be a continuous function and $r \in \mathbb{R}$. Show that $\text{st}(g^M(r + \varepsilon)) = g(r)$.

8. Let f be a function symbol corresponding to a differentiable function and let f' be the function symbol corresponding to its derivative. Show that for $r \in \mathbb{R}$ we have

$$\text{st} \left(\frac{f^M(r + \varepsilon) - f^M(r)}{\varepsilon} \right) = f'(r).$$

In particular, show that the term inside st is in $\langle \mathbb{R} \rangle$.

9. Prove the Leibniz rule $(fg)' = f'g + fg'$ using the above interpretation of the derivative.
10. Read the exercises below. Familiarise yourself with their statements. Their solutions are optional.

Fix a countable first-order language L . The set of L -formulas (in countably many variables) is sometimes denoted by $L_{\omega, \omega}$. This is to indicate that when we form L -formulas, we can only consider finite conjunctions $\bigwedge_{i=1}^n \varphi_i$ (with $n < \omega$) and finite quantifications $\exists x_1 \dots \exists x_m \varphi$ (with $m < \omega$). Given infinite cardinals κ, λ with $\lambda \leq \kappa$ and a set V of variables with $|V| = \kappa$ one can consider $L_{\kappa, \lambda}$ -preformulas which are defined similarly as $L_{\omega, \omega}$ -formulas (so we start with atomic formulas and have negation) but with the following connectives and quantification steps:

- If $\{\varphi_\alpha\}_{\alpha \in A}$ are $L_{\kappa, \lambda}$ -preformulas with $|A| < \kappa$ then so are

$$\bigwedge_{\alpha \in A} \varphi_\alpha, \bigvee_{\alpha \in A} \varphi_\alpha.$$

- If φ is a $L_{\kappa, \lambda}$ -preformula and $X \subset V$ satisfies $|X| < \lambda$, then so is $\exists X \varphi$.

We distinguish among those $L_{\kappa, \lambda}$ -formulas as those that have only fewer than λ free variables. In the following exercise we will be interested in $L_{\omega_1, \omega}$. In the proof system for $L_{\omega_1, \omega}$ we add axioms

$$\bigwedge_{\alpha \in A} \varphi_\alpha \rightarrow \varphi_{\alpha_0},$$

for any countable collection of formulas $\{\varphi_\alpha\}_{\alpha \in A}$ whose conjunction is a formula, and $\alpha_0 \in A$. Furthermore we add the deduction rule scheme

$$\frac{\varphi_0, \varphi_1, \dots}{\bigwedge_{i \in \mathbb{N}} \varphi_i},$$

for every sequence of $L_{\omega_1, \omega}$ -formulas $\{\varphi_i\}_{i \in \mathbb{N}}$ whose conjunction is a formula. Here proofs are infinitely branching trees with finite depth branches.

Exercise 2.18. © Imitate Henkin's proof to show that if an $L_{\omega_1, \omega}$ -sentence σ is consistent, then it has a model. Conclude with the completeness theorem for infinitary formulas $\models \varphi \iff \vdash \varphi$.

Some systems of axioms use more than one type of primitive object. For example, Euclid's axioms are about points P , lines l , and the incidence relation between them $P \in l$. One can formalise this in the framework of multisorted (first-order) logic, where a language L consists of:

- Sort symbols $\{S_\alpha\}_{\alpha \in A}$.
- Function symbols $\{f_i\}_{i \in I}$, each equipped with its arity, which is a tuple $(\alpha_1, \dots, \alpha_n; \beta)$ for some $n \in \mathbb{N}$ and $\alpha_1, \dots, \alpha_n, \beta \in A$.
- Predicate symbols $\{P_j\}_{j \in J}$, each equipped with its arity which is a tuple $(\alpha_1, \dots, \alpha_m)$ for some $m \in \mathbb{N}$ and $\alpha_1, \dots, \alpha_m \in A$.

- Constant symbols $\{c_k\}_{k \in K}$, each equipped with a sort of index $\alpha = \alpha(k) \in A$.

In that case, an L -structure M is a collection of non-empty sets S_α^M together with interpretations of other symbols, where if f is a function symbol with arity $(\alpha_1, \dots, \alpha_n; \beta)$, then its interpretation is a function

$$f^M : S_{\alpha_1}^M \times \dots \times S_{\alpha_n}^M \rightarrow S_\beta^M.$$

Similarly, if P is a predicate symbol with arity $(\alpha_1, \dots, \alpha_m)$, then its interpretation is a subset

$$P^M \subseteq S_{\alpha_1}^M \times \dots \times S_{\alpha_m}^M.$$

If c is a constant of sort S , then M is equipped with its interpretation $c^M \in S^M$. When defining L -formulas, we need to consider variables decorated by sort symbols so each sort has a different set of variables associated to it.

Exercise 2.19. (e) Convince yourself that Henkin's proof of completeness works for multisorted structures.

Exercise 2.20. (e) Look at Hilbert's axioms of Euclidean geometry and define an appropriate multisorted first-order language.

Many axiom systems involve a complete metric and so the real numbers in their definition. For example the axioms of Banach spaces or Hilbert spaces are such. Continuous (first-order) logic is a variant of first-order logic, where one can consider some real-valued predicates and the equality is replaced by the metric. In the following by a *continuity modulus* we mean a continuous increasing function

$$\delta : [0, 1] \rightarrow [0, 1]$$

satisfying $\delta(0) = 0$ and $\delta(\varepsilon) > 0$ for $\varepsilon > 0$.

Definition 2.21. A *continuous language* L consists of:

- sort symbols S each equipped with a metric symbol d_S and with a bound $B_S \in \mathbb{R}_{\geq 0}$;
- constant symbols c with their arity as in the multisorted logic;
- function symbols f with the same arities as in the multisorted logic, each equipped with its continuity modulus δ_f ;
- predicate symbols P with their arities same as in the multisorted logic, each equipped with its continuity modulus δ_P and with a bound $B_P \in \mathbb{R}_{\geq 0}$.

An L -structure M consists of interpretations of sorts, constant and function symbols, same as in multisorted logic, however, for each sort S its interpretation S^M is equipped with the metric symbol interpretation d_S^M which makes (S^M, d_S^M) a non-empty complete metric space of diameter at most B_S . Moreover, the function symbols interpretations are supposed to respect the continuity modulus in the following sense: for all real ε with $0 < \varepsilon \leq 1$, for every $a_1, \dots, a_n, b_1, \dots, b_n$ in appropriate sorts of M , we have

$$\max_{i=1, \dots, n} d(a_i, b_i) < \delta_f(\varepsilon) \implies d(f(a_1, \dots, a_n), f(b_1, \dots, b_n)) \leq \varepsilon,$$

where we skipped interpretation in M (and the indexes for sorts) in the notation. Furthermore, the interpretations of predicate symbols are functions

$$P^M : S_{\alpha_1}^M \times \dots \times S_{\alpha_m}^M \rightarrow [0, B_P]$$

that are required to respect the continuity modulus in the following sense: for all real $1 \geq \varepsilon > 0$ for every $a_1, \dots, a_m, b_1, \dots, b_m$ in appropriate sorts of M we have

$$\max_{i=1, \dots, m} d(a_i, b_i) < \delta_P(\varepsilon) \implies |P(a_1, \dots, a_m) - P(b_1, \dots, b_m)| \leq \varepsilon.$$

When we build formulas in continuous logic, terms are defined similarly as in the discrete case. More complicated expressions are constructed by:

- (i) atomic formulas - if t_1, t_2 are terms of the same sort S , then $d_S(t_1, t_2)$ is an atomic formula. If $t_1, \dots, t_n$ are terms of appropriate sorts, and P is a predicate symbol whose arity has matching sorts, then $P(t_1, \dots, t_n)$ is an atomic formula.
- (ii) connectives - if $\varphi_1, \dots, \varphi_n$ are formulas and $u : \mathbb{R}^n \rightarrow \mathbb{R}$ is a continuous function, then $u(\varphi_1, \dots, \varphi_n)$ is a formula.
- (iii) quantifiers - if φ is a formula then so are $\sup_x \varphi$ and $\inf_x \varphi$.

In some variants of continuous logic one restricts only to $[0, 1]$ -valued formulas (all the bounds are equal to 1), and connectives are taken to be only $x \mapsto 1 - x$, $x \mapsto x/2$, $(x, y) \mapsto \max(x - y, 0)$. If M is an L -structure, $\varphi(\bar{x})$ is a formula with free-variables $\bar{x}$ and $\bar{a} \in M^{\bar{x}}$, then one inductively defines $\varphi(\bar{a})^M \in \mathbb{R}$. A sentence is said to hold in M if it is evaluated to zero. It is possible to formalise the notion of a proof in continuous logic so that a completeness theorem holds, see [1]. Later, we may prove the following theorem, using ultraproducts.

Theorem 2.22 (Compactness for continuous logic). Let T be a continuous logic theory. Assume that for every $\varphi_1, \dots, \varphi_n \in T$ and every $\varepsilon > 0$ there is an L -structure M such that for all $i = 1, \dots, n$ we have $|\varphi_i^M| \leq \varepsilon$. Then T has a model.

Exercise 2.23. © Axiomatize the following structures in continuous logic: Banach spaces, Hilbert spaces, C^* -algebras. Hint: introduce a sort for each ball of radius $n \in \mathbb{N}$.

3. LÖWENHEIM–SKOLEM THEOREM

Fix a first-order language L , so if we refer to formulas or sentences we mean the ones over L . We use letters M, N to denote L -structures.

Definition 3.1. (1) A *homomorphism between M and N* is a function $F : M \rightarrow N$ such that for every function symbol f of arity n , predicate symbol P of arity m , and constant symbol c we have

$$\begin{cases} F(f^M(a_1, \dots, a_n)) = f^N(F(a_1), \dots, F(a_n)), \\ P^M(b_1, \dots, b_m) \implies P^N(F(b_1), \dots, F(b_m)), \\ F(c^M) = c^N, \end{cases}$$

for any a_i 's and b_j 's in M . In particular, L -structures form a category, and an *isomorphism* is a homomorphism that has an inverse homomorphism. We write $M \cong N$ to denote that M and N are isomorphic L -structures.

- (2) We say that M is a *substructure of N* , written $M \subseteq N$, if corresponding sets are contained in each other, and interpretations of all symbols in N restrict to interpretations of these symbols in M .
- (3) We say that M is *elementarily equivalent to N* , written $M \equiv N$, if for every sentence φ we have $M \models \varphi \iff N \models \varphi$.

- (4) We say that M is an *elementary substructure* of N , written $M \prec N$, if M is a substructure of N and for any finite tuple $\bar{a} \subseteq M$ and any formula $\varphi(\bar{x})$ with $|\bar{x}| = |\bar{a}|$ we have

$$M \models \varphi(\bar{a}) \iff N \models \varphi(\bar{a}).$$

- (5) A function $\alpha : M \rightarrow N$ is *elementary* if for any finite tuple $a_1, \dots, a_n \in M$ and any formula $\varphi(\bar{x})$ with $|\bar{x}| = |\bar{a}|$ we have

$$M \models \varphi(a_1, \dots, a_n) \iff N \models \varphi(\alpha(a_1), \dots, \alpha(a_n)).$$

Theorem 3.2 (Tarski-Vaught test). Let A be a subset of N . Then A is a universe of an elementary substructure of N if and only the following condition $(\star)$ holds: For every formula $\varphi(x, \bar{y})$ and every $\bar{a} \in A^{|\bar{y}|}$ if $N \models \exists x \varphi(x, \bar{a})$, then there is $b \in A$ such that $N \models \varphi(b, \bar{a})$.

Proof. First, we assume that A is a universe of an elementary substructure. Then by the assumption

$$N \models \exists x \varphi(x, \bar{a}) \iff A \models \exists x \varphi(x, \bar{a}),$$

for $\varphi(x, \bar{y})$ and $\bar{a}$ as in $(\star)$, so there is $b \in A$ such that $A \models \varphi(b, \bar{a})$.

Now assume that $(\star)$ holds. Note that this implies that A is closed under N -interpretations of function symbols, as for a function symbol f of arity n , if $a_1, \dots, a_n \in A$, then by $(\star)$ we get

$$N \models \exists x f(a_1, \dots, a_n) = x \implies \text{there is } b \in A \text{ such that } f(a_1, \dots, a_n) = b.$$

Now we show that $A \prec N$ inductively, based on the complexity of the formula. Atomic formulas and connective steps are easy $\textcircled{e}$. Consider the formula $\varphi(\bar{y}) = \exists x \psi(x, \bar{y})$ and assume that ψ satisfies the inductive hypothesis. Let $\bar{a} \in A^{|\bar{y}|}$. We get

$$\begin{aligned} N \models \exists x \psi(x, \bar{a}) &\implies N \models \psi(b, \bar{a}) \text{ for some } b \in A \\ &\iff A \models \psi(b, \bar{a}) \text{ for some } b \in A \iff A \models \exists x \psi(x, \bar{a}), \end{aligned}$$

where the first implication follows by $(\star)$ and is actually an equivalence, and the middle equivalence follows from the inductive hypothesis. $\square$

By convention, the cardinality of the language is the cardinality of the set of formulas in countably many variables. Thus, it is maximum of the cardinality of the set of symbols and $\aleph_0$.

Corollary 3.3 (Löwenheim-Skolem theorem). Let M be an infinite L -structure and assume that $|L| = \kappa$. Then:

- a) There is an $N \prec M$ with $|N| \leq \kappa$.
- b) For any $\lambda \geq \max(\kappa, |M|)$ there is an L -structure N with $|N| = \lambda$ and with an elementary embedding $F : M \rightarrow N$.

Proof. a) Let $(\varphi_\alpha(x, \bar{y}))_{\alpha < \kappa}$ be an enumeration of all L -formulas where each formula appears cofinally many times. We define a sequence of subsets $(A_\alpha)_{\alpha < \kappa}$ inductively with $A_0 = \emptyset$ and A_β defined so that it contains $\bigcup_{\alpha < \beta} A_\alpha$ and for every $\alpha < \beta$ and every $\bar{a} \subseteq A_\alpha$ of the same length as $\bar{y}$ in $\varphi_\alpha(x, \bar{y})$, if $M \models \exists x \varphi_\alpha(x, \bar{a})$, then we choose a witness and add it to A_β . If $N = \bigcup_{\alpha < \kappa} A_\alpha$, then $|N| \leq \kappa$ and it satisfies the Tarski-Vaught test, so $N \prec M$.

- b) Let $L' = L \cup \{c_\alpha : \alpha \in M\} \cup \{d_\alpha : \alpha < \lambda\}$ and consider the L' -theory

$$T' = \{\varphi(c_{a_1}, \dots, c_{a_n}) : \text{if } M \models \varphi(a_1, \dots, a_n)\} \cup \{d_\alpha \neq d_\beta : \alpha \neq \beta < \lambda\}.$$

First note that this theory is consistent. Indeed, any finite part of T' uses only finitely many axioms of the form $d_\alpha \neq d_\beta$. Thus if we interpret $c_a^M = a$ and d_α 's appearing in the finite fragment as any distinct elements (here we use the infinitude of M), we get consistency of that part. Using the completeness theorem, T' has a model. Either by our proof of completeness, or by a), we get that T' has a model N of cardinality exactly λ . We define $F : M \rightarrow N$ by $F(a) = c_a^N$. Because of the axioms $\varphi(c_{a_1}, \dots, c_{a_n})$ in T' , we get that F is an elementary embedding. $\square$

Note that in the upper Löwenheim–Skolem theorem we can construct $M \prec N'$, with N' of cardinality λ , by extending $F : M \rightarrow N$ to a bijection $F' : N' \rightarrow N$ (where N' is some set of cardinality λ containing M) and transporting back the L -structure from N to N' via the bijection F' .

For an L -structure M , we denote by $Th(M)$ the set of L -sentences true in M - we call it the *complete theory* of M . If $A \subseteq M$ is a subset (that can be the whole M) and $L' = L \cup \{c_a : a \in A\}$ and

$$\text{Diag}_{\text{el}}(A) = \{\varphi(c_{a_1}, \dots, c_{a_n}) : \text{if } M \models \varphi(a_1, \dots, a_n)\}$$

we call $\text{Diag}_{\text{el}}(A)$ the *elementary diagram* of A . Similarly, we call

$$\text{Diag}(A) = \{\varphi(c_{a_1}, \dots, c_{a_n}) : \text{if } M \models \varphi(a_1, \dots, a_n) \text{ and } \varphi \text{ is } (\neg)\text{atomic}\}$$

the *atomic diagram* of A (we include sentences that come from atomic formulas and their negations). We call a formula φ *quantifier-free* if in its inductive definition no quantifiers are used. Also, we call a formula *universal* (resp. *existential*) if it is of the form $\forall \bar{x} \varphi$ (resp. $\exists \bar{x} \varphi$) for some quantifier-free formula φ . We call a formula an $\forall\exists$ -formula if it is of the form $\forall \bar{x} \exists \bar{y} \varphi$, where φ is quantifier-free.

Exercises, sheet 2. Fix a first-order language L . M and N will denote L -structures and T will denote an L -theory. If $(I, <)$ is a linear order, by a (an elementary) chain of L -structures we mean a collection $(M_i)_{i \in I}$ such that $M_i \subseteq M_j$ if $i < j$ (resp. $M_i \prec M_j$).

1. Show that any for any finite structure in a finite language there is a sentence that determines it up to an isomorphism.
2. Show that isomorphic structures satisfy the same sentences. Hint: induction based on the complexity of the formula.
3. Show for an elementary chain $(M_i)_{i \in I}$, each M_i is elementarily embedded in the union of all M_j 's (define a natural L -structure on this union). Generalise to the case where $(I, <)$ is a directed system.
4. For a theory T let $T_\forall$ be the set of universal sentences that are consequences of T . Show that $M \models T_\forall$ if and only if there is an $N \models T$ with $M \subseteq N$.
5. Deduce that a formula $\varphi(\bar{x})$ is T -equivalent to a universal formula if and only if for every $N \subseteq M$, $M, N \models T$ and $\bar{a} \in N^{|\bar{x}|}$, we have

$$M \models \varphi(\bar{a}) \implies N \models \varphi(\bar{a}).$$

6. Show that $\text{Diag}_{\text{el}}(M)$ (resp. $\text{Diag}(M)$) is consistent with T , then there is an elementary embedding (resp. an embedding of L -structures) of M in a model of T .
7. We call T *model complete* if for every model $M \models T$, the $L(M)$ -theory $T \cup \text{Diag}(M)$ is complete (where $L(M) = L \cup \{c_m : m \in M\}$). Show that the following are equivalent:
 - T is model complete,

- every embedding of models of T is elementary,
 - every formula is T -equivalent to a universal one,
 - every formula is T -equivalent to an existential one.
8. Show that a theory T can be axiomatized by $\forall\exists$ -formulas (i.e., T is *inductive*) if and only if for any chain of L -structures $(M_i)_{i \in I}$ that are models of T , their union is also a model of T .
9. A model $M \models T$ is called *existentially closed* if for any extension $M \subseteq N$ with $N \models T$, M is *existentially closed in N* (written $M \prec_{\exists} N$), i.e., whenever $\varphi(\bar{y})$ is an existential formula and $\bar{b} \in M^{|\bar{y}|}$, then

$$M \models \varphi(\bar{b}) \iff N \models \varphi(\bar{b}).$$

Show that if T is inductive, then any model of T can be embedded in an existentially closed model of T . Show that T is model complete if and only if all of its models are existentially closed.

10. An L -theory T^* is a *model companion* of T if:
- every model of T admits an embedding into a model of T^* ,
 - every model of T^* admits an embedding into a model of T ,
 - T^* is model complete.
- Show that if a model companion of T exists, it is unique up to equivalence of theories.
11. Show that if T is inductive, its model companion exists if and only if the class of its existentially closed models is axiomatized by a theory (i.e., this class is *elementary*).

4. CRITERIA FOR ELEMENTARY EQUIVALENCE

Fix a language L and let M, N be L -structures.

Question 4.1. How to check if $M \equiv N$?

There is a criterion for elementary equivalence in form of a game. Assume for simplicity that L is finite and has only relation symbols. It is a minor restriction as we can encode function symbols by their graphs and constants by new unary predicate symbols.

Definition 4.2. The *Ehrenfeucht–Fraïssé game of length $n \in \mathbb{N}$* , denoted by $\Gamma_n(M, N)$, is being played between two players I and II. The first player is called Spoiler and the second is called Duplicator.

- The game consists of rounds indexed by $i = 1, \dots, n$. At the i 'th round, elements $a_1, \dots, a_{i-1} \in M$ and $b_1, \dots, b_{i-1} \in N$ are already chosen.
- The player I does one of the following two things:
 - (1) either chooses $a_i \in M$, which is answered by a choice of $b_i \in N$ by the player II;
 - (2) or chooses $b_i \in N$, for which the player II answers with a choice of $a_i \in M$.
- At the end of the game the player II wins, if the resulting sequences $(a_1, \dots, a_n) \in M^n$ and $(b_1, \dots, b_n) \in N^n$ are such that the map $a_i \mapsto b_i$ is well defined and is a *partial isomorphism* of M and N , i.e., an isomorphism between L -substructures consisting of these n -tuples.

Example 4.3. Consider two linear orders: $(\mathbb{Z}, <)$ and $(\mathbb{Z} \sqcup \mathbb{Z}, <)$ (one copy after another). By playing “far enough” one can show ⑥ that II has a winning strategy in the length n Ehrenfeucht–Fraïssé game, for every $n \in \mathbb{N}$.

Theorem 4.4 (Ehrenfeucht–Fraïssé). M and N are elementarily equivalent if and only if II has a winning strategy in $\Gamma_n(M, N)$ for every $n \in \mathbb{N}$.

Proof. ⑥ Exercise. □

Let us come back to an arbitrary language L and an L -theory T .

Definition 4.5. Let κ be an infinite cardinal. A theory T is κ -categorical if it has a model of cardinality κ and for any two models $M, N \models T$ of cardinality κ , $M \cong N$.

Here is another criterion for elementary equivalence. Recall that due to our convention of identifying a theory with a set of its logical consequences, T is complete if for every sentence φ , $T \vdash \varphi$ or $T \vdash \neg\varphi$.

Theorem 4.6 (Łoś–Vaught test). If T has no finite models and is κ -categorical, for some infinite cardinal κ bigger or equal the cardinality of L , then T is complete.

Proof. Assume for a contradiction that there is a sentence φ such that $T \not\vdash \varphi$ and $T \not\vdash \neg\varphi$. By completeness (and deduction), there are $M \models T \cup \{\neg\varphi\}$ and $N \models T \cup \{\varphi\}$ of the cardinality of L . By the (upper) Löwenheim–Skolem theorem we can find elementary extensions $M \prec M', N \prec N'$ of cardinality κ . Using κ -categoricity we get that $M' \cong N'$. Hence

$$M \equiv M' \equiv N' \equiv N$$

and we get a contradiction. □

Example 4.7. Consider the theory DLO_0 of dense linear orders without endpoints. It is a theory in the language $L = \{<\}$ axiomatized by the usual linear order axioms together with

$$(\exists x)(\exists y)(x < y) \wedge (\forall x, y)(x < y \rightarrow (\exists w, u, z)(w < x < u < y < z)).$$

Consider two countable models $M, N \models \text{DLO}_0$. We will show that $M \cong N$ (Cantor’s theorem). Enumerate $M = \{a_n : n \in \mathbb{N}\}, N = \{b_n : n \in \mathbb{N}\}$. We construct inductively a sequence of partial isomorphisms f_n , where $A_n \subseteq M, B_n \subseteq N$ are sequences of finite subsets such that

- i. $A_n \subseteq A_{n+1}, B_n \subseteq B_{n+1}, f_n \subseteq f_{n+1}, A_0 = B_0 = f_0 = \emptyset$,
- ii. $f_n : A_n \rightarrow B_n$ is an isomorphism of induced finite orders,
- iii. $a_n \in A_{2n+1}, b_n \in B_{2n+2}$.

The construction is inductive. Assume we have f_{2n}, A_{2n}, B_{2n} . If $a_n \in A_{2n}$, we put $(f_{2n+1}, A_{2n+1}, B_{2n+1}) = (f_{2n}, A_{2n}, B_{2n})$. Otherwise we write elements of A_{2n} as

$$c_1 < \dots < c_{i-1} < a_n < c_i < \dots < c_k,$$

and we find $b \in N$ such that $f_{2n}(c_{i-1}) < b < f_{2n}(c_i)$ (if a_n is the biggest or the smallest element we choose b appropriately). Then we put $f_{2n+1} = f_{2n} \cup \{(a_n, b)\}$ and $A_{2n+1} = A_{2n} \cup \{a_n\}, B_{2n+1} = B_{2n} \cup \{b\}$. On the odd steps we reverse the construction. If we put $f = \bigcup_{n \in \mathbb{N}} f_n$ we get an isomorphism $f : M \rightarrow N$. In particular DLO_0 is complete.

Corollary 4.8. DLO_0 is decidable. More precisely, there is an algorithm that produces sentences true in any model of DLO_0 , such that for any sentence σ , it will produce σ or $\neg\sigma$ at some point.

Proof. Note that DLO_0 is given by a list of finitely many axioms. If our program lists all formal proofs that use this list of axioms, by completeness of DLO_0 we know that it has the desired property. $\square$

In the above proof, it would be enough to know that DLO_0 is recursively axiomatizable, i.e., that there exists a computer program listing its axioms (and possibly working till the end of time).

Another tool in determining elementary equivalence of L -structures is quantifier-elimination.

Definition 4.9. We say that an L -theory T *eliminates quantifiers* if for every L -formula φ there is a quantifier-free formula ψ such that $T \vdash \varphi \leftrightarrow \psi$.

We start with an example. Let $L = \{E\}$ where E is a binary relation symbol and let T be the theory saying that E is an equivalence relation with infinitely many infinite classes. We will prove directly that it has quantifier elimination. Consider a quantifier-free formula

$$\psi(\bar{x}, y) = \bigvee_{i=1}^n \bigwedge_{j=1}^{J_i} (\alpha_{ij} F_{ij} \beta_{ij}),$$

where $\alpha_{ij}, \beta_{ij} \in \{x_1, \dots, x_n, y\}$ and $F_{ij} \in \{=, \neq, E, \neg E\}$. We will show that $\exists y \psi(\bar{x}, y)$ is T -equivalent to a quantifier-free formula in variables $\bar{x}$. Without loss of generality we can assume that

$$\psi(\bar{x}, y) = \bigwedge_{j=1}^m (\alpha_j F_j \beta_j).$$

Split this conjunction into terms that do not contain y and the other ones, so that we can assume that, up to rearranging and using the symmetry of relations considered, we have

$$\psi(\bar{x}, y) = \bigwedge_{j=1}^k (x_{i_j} F_j x_{i'_j}) \wedge \bigwedge_{l=k+1}^m (y F_l x_{i_l}).$$

If one of the relations $y F_l x_{i_l}$ is $y = x_{i_l}$, then $\exists y \psi(\bar{x}, y)$ is equivalent to $\psi(\bar{x}, x_{i_l})$. Otherwise, assume that the new sentences say that

$$y \neq x_{j_1}, \dots, y \neq x_{j_a}, y E x_{j_{a+1}}, \dots, y E x_{j_b}, \neg y E x_{j_{b+1}}, \dots, \neg y E x_{j_p}.$$

If $a = b$, then in a model of T , $\exists y \psi(\bar{x}, y)$ holds for a tuple $\bar{c}$ if and only if $\bar{c}$ satisfies

$$\bigwedge_{j=1}^k (x_{i_j} F_j x_{i'_j}).$$

If $a < b$, then in a model of T , $\exists y \psi(\bar{x}, y)$ holds for a tuple $\bar{c}$ if and only if $\bar{c}$ satisfies

$$\bigwedge_{j=1}^k (x_{i_j} F_j x_{i'_j}) \wedge \bigwedge_{l=a+1}^{b-1} (x_{j_l} E x_{j_b}) \wedge \bigwedge_{l=b+1}^p (\neg x_{j_b} E x_{j_l}).$$

Thus we showed that T has quantifier-elimination. Note that the same algorithm works, if we added two constants to L , say $L' = L \cup \{c, d\}$. However, with two constants the theory $T' = T$ (but in L') is not complete, as it does not determine whether $c E d$ or whether $c = d$. Quantifier elimination for T' implies that every L' -sentence is T' -equivalent to a Boolean combination of sentences $c E d, c = d$. Thus determining these relations is the only obstruction to completeness of T' . Later we will develop more conceptual criteria for quantifier-elimination. To do that we need to introduce types.

5. STONE DUALITY AND ULTRAPRODUCTS

Fix a theory T in L . We say that two formulas $\varphi(\bar{x}), \psi(\bar{x})$ are *equivalent* (modulo T) if $T \vdash \varphi \leftrightarrow \psi$. Equivalence classes of formulas form a *Boolean algebra*, i.e., we can define operations $\wedge, \vee, \neg$ on them, and the usual laws of set operations $\cap, \cup, (-)^c$ are satisfied ($\wedge, \vee$ are associative, idempotent, commutative, and distributive with respect to each other; $\neg$ is involutive, satisfies the law of excluded middle, and De Morgan's laws hold). Furthermore, this algebra has $0 = [\bigwedge_i x_i \neq x_i]$ and $1 = [\bigwedge_i x_i = x_i]$.

Definition 5.1. Fix a tuple of variables $\bar{x} = (x_1, \dots, x_n)$. The *Lindenbaum-Tarski algebra* is the Boolean algebra of T -equivalence classes of formulas in variables $\bar{x}$, where we define $[\varphi] \wedge [\psi] = [\varphi \wedge \psi]$, etc.

Definition 5.2. A *type* in variables $\bar{x} = (x_1, \dots, x_n)$ is a collection of formulas

$$p(\bar{x}) = \{\varphi_i(\bar{x}) : i \in I\}$$

such that every finite subset $p_0(\bar{x}) \subseteq p(\bar{x})$, $T \cup \{\exists \bar{x} \bigwedge p_0(\bar{x})\}$ is consistent. We call p a *complete type*, if for any formula φ in variables $\bar{x}$, either $\varphi \in p$ or $\neg\varphi \in p$.

Non-complete (or not necessarily complete) types are sometimes called *partial types*.

Exercise 5.3. (e) Let $p(\bar{x})$ be a type with respect to T . Show that there is a model $M \models T$ and a tuple $\bar{a} \in M^{|\bar{x}|}$ such that for every $\psi \in p$ we have $M \models \psi(\bar{a})$.

Let $(B, \wedge, \vee, \neg, 0, 1)$ be a Boolean algebra. A (proper) *filter* is a subset $F \subseteq B$ such that ($0 \notin F$ and):

- $1 \in F$,
- $a \in F \ \& \ \neg a \vee b = 1 \implies b \in F$,
- $a, a' \in F \implies a \wedge a' \in F$.

In the second bullet the relation $\neg a \vee b = 1$ is sometimes denoted by $a \subset b$.

Exercise 5.4. (e) Define $a \triangle b = (a \wedge \neg b) \vee (b \wedge \neg a)$. Show that B with $+$:= $\triangle$, $\cdot$:= $\wedge$ is a commutative ring with $0, 1$ being respectively the neutral elements of addition and multiplication. Show that an $F \subseteq B$ is a proper filter if and only if $\{\neg b : b \in F\}$ is a proper ideal.

A proper filter is called an *ultrafilter* if it is maximal with respect to inclusion of proper filters. In particular, $F \subseteq B$ is an ultrafilter if and only if $\{\neg b : b \in F\}$ is a maximal proper ideal in $(B, +, \cdot)$. A proper ideal in $(B, +, \cdot)$ is maximal if and only if it is prime; the quotient by such an ideal is the two-element field (e).

Exercise 5.5. (e) A filter $F \subseteq B$ is an ultrafilter if and only if for every $b \in B$, exactly one of the two possibilities happen: $b \in F$ or $\neg b \in F$.

Definition 5.6. The *Stone space* $\text{Stone}(B)$ of a Boolean algebra B is the set of ultrafilters on B equipped with the topology where the basis open sets are (finite intersections of) sets of the form

$$[b] := \{F \in \text{Stone}(B) : b \in F\},$$

where we vary $b \in B$. Equivalently, we could define $\text{Stone}(B)$ as the set of prime ideals in $(B, +, \cdot)$, equipped with the Zariski topology ($\textcircled{e}$ for those who know some algebraic geometry).

For a tuple of variables $\bar{x} = (x_1, \dots, x_n)$ we denote by $S_n(T)$ (or $S_{\bar{x}}(T)$) the Stone space of the Lindenbaum-Tarski algebra in $\bar{x}$. We call it the space of *complete n -types*.

Note that we do not need to consider finite intersections of sets of the form $[b]$ as $[b_1] \cap [b_2] = [b_1 \wedge b_2] \textcircled{e}$.

Formally, an element of $S_n(T)$ is an ultrafilter of T -equivalence classes of L -formulas in variables $x_1, \dots, x_n$. We will be identifying such an element with the corresponding complete type consisting of all representatives of these equivalence classes.

Proposition 5.7. Stone space of any Boolean algebra B is compact, Hausdorff, and has a basis consisting of clopen sets.

Proof. First note that $[b]_{b \in B}$ is a basis of clopen subsets as $[b]^c = [\neg b]$. If $F_1, F_2 \subseteq B$ are two different ultrafilters in B , then there is $b \in B$ such that $b \in F_1$ and $\neg b \in F_2$, so that $F_1 \in [b]$ & $F_2 \in [b]^c = [\neg b]$. This shows Hausdorffness.

For compactness, assume that $[b_i]_{i \in I}$ is a centered family. This means that for every finite $I_0 \subseteq I$ the set $\bigcap_{i \in I_0} [b_i]$ is non-empty, i.e., the ideal generated by $(\neg b_i : i \in I_0)$ in $(B, +, \cdot)$ is a proper ideal. Thus, the ideal generated by all $\neg b_i$'s is a proper ideal, and any maximal ideal containing it will witness $\bigcap_{i \in I} [b_i] \neq \emptyset$. $\square$

A topological space satisfying the conditions from the preceding proposition is called a *Stone space*. The set of clopen subsets of a Stone space X is a Boolean algebra $\text{Bool}(X)$.

Theorem 5.8 (Stone duality). There is a correspondence between Boolean algebras and Stone spaces. More precisely, $\text{Bool}(\text{Stone}(B)) \cong B$, $\text{Stone}(\text{Bool}(X)) \approx X$ for any Boolean algebra B and any Stone space X .

Proof. Exercise $\textcircled{e}$. $\square$

For those who already know category theory (we will talk about it soon) - Stone duality is an (anti-)equivalence of categories when we consider continuous maps between Stone spaces and homomorphisms between Boolean rings.

If I is a set and $\mathcal{U} \subseteq P(I)$ is an ultrafilter on the Boolean algebra of all subsets of I , we can identify $\mathcal{U}$ with a finitely additive probabilistic measure $\mu : P(I) \rightarrow \{0, 1\}$, where $\mu(A) = 1 \iff A \in \mathcal{U}$.

Definition 5.9. Let $(M_i)_{i \in I}$ be a collection of L -structures and let $\mathcal{U}$ be an ultrafilter on $P(I)$. The *ultraproduct of $(M_i)_{i \in I}$ with respect to $\mathcal{U}$* , denoted $\prod_{\mathcal{U}} M_i$ or $\prod_{i \in I} M_i / \mathcal{U}$, is the following L -structure:

- (i) The universe of $\prod_{i \in I} M_i / \mathcal{U}$ is the set of equivalence classes of sequences in $\prod_{i \in I} M_i$ where we identify two sequences that are equal 'almost everywhere'

with respect to the measure μ . We denote the class of $(a_i) \in \prod_{i \in I} M_i$ by $[(a_i)]_{\mathcal{U}}$.

- (ii) If f is a function symbol of arity n , then we define its interpretation in $\prod_{i \in I} M_i/\mathcal{U}$ to be:

$$f([(a_i^1)]_{\mathcal{U}}, \dots, [(a_i^n)]_{\mathcal{U}}) = [(f(a_i^1, \dots, a_i^n))]_{\mathcal{U}}.$$

- (iii) If P is a predicate symbol of arity m , then we define its interpretation in $\prod_{i \in I} M_i/\mathcal{U}$ to be:

$$P([(b_i^1)]_{\mathcal{U}}, \dots, [(b_i^m)]_{\mathcal{U}}) \iff \{i \in I : M_i \models P(b_i^1, \dots, b_i^m)\} \in \mathcal{U},$$

- (iv) If c is a constant symbol, then we define its interpretation in $\prod_{i \in I} M_i/\mathcal{U}$ to be $[(c^{M_i})]_{\mathcal{U}}$.

We leave it to the reader (©) to verify that in this definition of L -structure function and predicate symbols interpretations only depend on elements of $\prod_{i \in I} M_i/\mathcal{U}$ and not of their representatives in $\prod_{i \in I} M_i$.

Theorem 5.10 (Łoś's theorem). Let $\varphi(x_1, \dots, x_n)$ be an L -formula. Then

$$\prod_{i \in I} M_i/\mathcal{U} \models \varphi([(a_i^1)]_{\mathcal{U}}, \dots, [(a_i^n)]_{\mathcal{U}}) \iff \{i \in I : M_i \models \varphi(a_i^1, \dots, a_i^n)\} \in \mathcal{U}.$$

Proof. We do it inductively based on the complexity of φ . The case of atomic formulas and connectives is left as an exercise (©). Let $\varphi(\bar{x}) = \exists y \psi(\bar{x}, y)$. First, assume that $I_0 = \{i \in I : M_i \models \varphi(a_i^1, \dots, a_i^n)\} \in \mathcal{U}$. This means that for $i \in I_0$ we can find $b_i \in M_i$ such that $M_i \models \psi(a_i^1, \dots, a_i^n, b_i)$. Define $(b_i) \in \prod_{i \in I} M_i$ arbitrarily outside of I_0 . By the inductive hypothesis we have

$$\prod_{i \in I} M_i/\mathcal{U} \models \psi([(a_i^1)]_{\mathcal{U}}, \dots, [(a_i^n)]_{\mathcal{U}}, [(b_i)]_{\mathcal{U}}),$$

so by definition the left hand side of the desired equivalence holds.

Now, let us assume that

$$\prod_{i \in I} M_i/\mathcal{U} \models \varphi([(a_i^1)]_{\mathcal{U}}, \dots, [(a_i^n)]_{\mathcal{U}}).$$

Let $[(b_i)]_{\mathcal{U}}$ be such that $\prod_{i \in I} M_i/\mathcal{U} \models \psi([(a_i^1)]_{\mathcal{U}}, \dots, [(a_i^n)]_{\mathcal{U}}, [(b_i)]_{\mathcal{U}})$. By the inductive hypothesis, this means that $\{i \in I : M_i \models \psi(a_i^1, \dots, a_i^n, b_i)\} \in \mathcal{U}$. However this set is contained in $\{i \in I : M_i \models \exists y \psi(a_i^1, \dots, a_i^n, y)\}$, so we get that the latter also is in $\mathcal{U}$. $\square$

An ultrafilter $\mathcal{U} \subseteq P(I)$ is called *principal*, if it is of the form

$$\mathcal{U} = \{J \subseteq I : i_0 \in J\}$$

for some $i_0 \in I$. In particular, an ultrafilter is non-principal if and only if it contains the filter of all cofinite subsets of I (©). Non-principal ultrafilter are very useful as the ultraproducts with respect to them give a way to ‘average’ an infinite family of structures into a single one.

6. REALISING AND OMITTING TYPES

Fix an L -theory T .

Definition 6.1. Let $A \subseteq M$ be a subset of $M \models T$. We write $L(A) = L \cup \{c_a : a \in A\}$ for the expansion of the language by copies of elements of A . For $\bar{b} \in M^{|\bar{x}|}$ we write $\text{tp}(\bar{b}/A)$ for the set of equivalence classes of formulas $\varphi(\bar{x}) \in L(A)$ such that $M \models \varphi(\bar{b})$. We denote by $S_{\bar{x}}(A)$ the space of complete $\bar{x}$ -types relative to $\text{Diag}_{\text{el}}(A)$ as an $L(A)$ -theory. In particular $\text{tp}(\bar{b}/A) \in S_{\bar{x}}(A)$.

Note that a type $p \in S_{\bar{x}}(A)$ carries information about the complete theory $\text{Th}(M)$ (even $\text{Diag}_{\text{el}}(A)$) that extends T . Indeed, this is because sentences can be treated as formulas in variables $\bar{x}$ (e.g. by conjugating $\sigma \wedge \bigwedge_{i=1}^n (x_i = x_i)$ for an $L(A)$ -sentence σ). For this reason, type spaces over parameter sets are typically considered when one is interested in a complete theory (or in a specific structure and its elementary extensions). In that case, when a complete theory T is clear from the context, we write $S_n(\emptyset)$ for $S_n(T)$.

Lemma 6.2. Consider a type $p \in S_n(A)$. There is an elementary extension $M \prec N$ and $\bar{b} \in N^n$ such that $p = \text{tp}(\bar{b}/A)$.

Proof. Consider the theory $T' = \text{Diag}_{\text{el}}(M) \cup p(\bar{x})$ in the language $L' = L(M) \cup \{x_1, \dots, x_n\}$ (so that we now treat m 's from M and x_i 's as constants). Choose a finite $p_0 \subseteq p$. Then, by definition

$$(m)_{m \in M} \models \text{Diag}_{\text{el}}(M) \text{ and } M \models \exists \bar{x} \bigwedge p_0(\bar{x}),$$

so if $\bar{c} \in M^n$ is a witness for the latter, then M with the natural $L(M)$ -structure, and with $x_i^M = c_i$ shows that $\text{Diag}_{\text{el}}(M) \cup p_0(\bar{x})$ is consistent. Since p_0 was arbitrary finite subset of p , we are done by compactness. $\square$

Since now on we allow ourselves to be a bit sloppy about extending the language by constants as in the above proof, in the sense that sometimes we will be considering variables or elements of a model as new constant symbols.

Exercises, sheet 3.

1. Prove the Ehrenfeucht–Fraïssé theorem.
2. Prove that the following $L = \{E\}$ -theories have quantifier elimination:
 - E is an equivalence relation with infinitely many equivalence classes of cardinality 7,
 - E is an equivalence relation with 7 infinite equivalence classes.
3. Solve all small exercises about Boolean algebras.
4. Prove that a subset $X \subseteq S_n(T)$ is clopen if and only if it is of the form $[\varphi]$ for some formula φ in n variables. Characterise closed subsets of $S_n(T)$. What about $\text{Stone}(B)$ for a Boolean algebra B ?
5. Prove that a complete type $p \in S_n(T)$ is isolated if and only if $\{p\}$ is a clopen set in $S_n(T)$.
6. Use ultraproducts to give another proof of the compactness theorem (this proof generalises easily to continuous logic).
7. Let $p(\bar{x})$ be a countable collection of formulas in variables $\bar{x}$ and assume that M_i are L -structures such that p is a type with respect to $\text{Th}(M_i)$, for $i \in \mathbb{N}$. Let $\mathcal{U}$ be a non-principal ultrafilter on $\mathbb{N}$. Show that p is realised in $\prod_{i \in \mathbb{N}} M_i / \mathcal{U}$.

8. Prove that $|S_n(A)| \leq 2^{|A|+|L|}$.

The following is an optional exercise for fans of descriptive set theory. Let T be a countable theory in a countable language L . Consider

$$X = \prod_f \omega^{\omega^{n_f}} \times \prod_P 2^{\omega^{m_P}} \times \prod_c \omega,$$

where the products are taken over the sets of function, predicate and constant symbols in L respectively. Let $\text{Mod}_\omega(T) \subseteq X$ be the subset consisting of L -structures on ω that are models of T . Define a topology on X with basic open subsets of the form

$$\{M \in X : M \models \varphi(\bar{a})\},$$

where $\varphi(\bar{x})$ varies over L -formulas and $\bar{a} \in \omega^{|\bar{x}|}$. Prove that:

- X is a Polish space.
- $\text{Mod}_\omega(T) \subseteq X$ is a closed subset, hence also a Polish space.
- If a partial type p is not isolated, then the set $\{M \in \text{Mod}_\omega(T) : M \text{ omits } p\}$ is a dense G_δ in $\text{Mod}_\omega(T)$.
- For a countable collection of non-isolated types $p_1, p_2, p_3, \dots$, there is a countable model $M \models T$ that omits all of them.

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